

Warranty Predictions

SoH-to-warranty survival & km-limit exposure

Mahindra · Euler · Bajaj | 22 Jun 2026

Scope

For every vehicle we project SoH forward to its warranty boundary and flag whether it falls below the 80% end-of-first-life threshold before expiry — and separately whether the km limit binds first. Results are split by model/term, because warranty exposure is term-dependent, not a single number.

Methodology

How warranty survival is predicted

Prediction

- Free-run the per-OEM degradation model from each vehicle's last observation to its warranty boundary.
- Survive = projected SoH stays $\geq 80\%$ (end-of-first-life) through the warranty term.
- Status tiers: OK $\geq 80\%$ · WATCH 78–80% (within forecast tolerance) · AT-RISK $< 78\%$.

Two limits, whichever binds first

- Time: warranty years from registration (Mahindra Treo 3 yr / Zor Grand 5 yr; Euler 5 yr; Bajaj 3 yr).
- Distance: the km cap (120,000 km). We check which limit a vehicle hits first.
- Mahindra is low-mileage (time binds); Bajaj is high-mileage (km binds).

Confidence

- Vehicles with < 8 months of history, or a warranty horizon $> 2 \times$ current age, are low-confidence.
- Two 'Ape E-Xtra FX' (a Piaggio model mislabelled into the Mahindra feed) are excluded from the split.

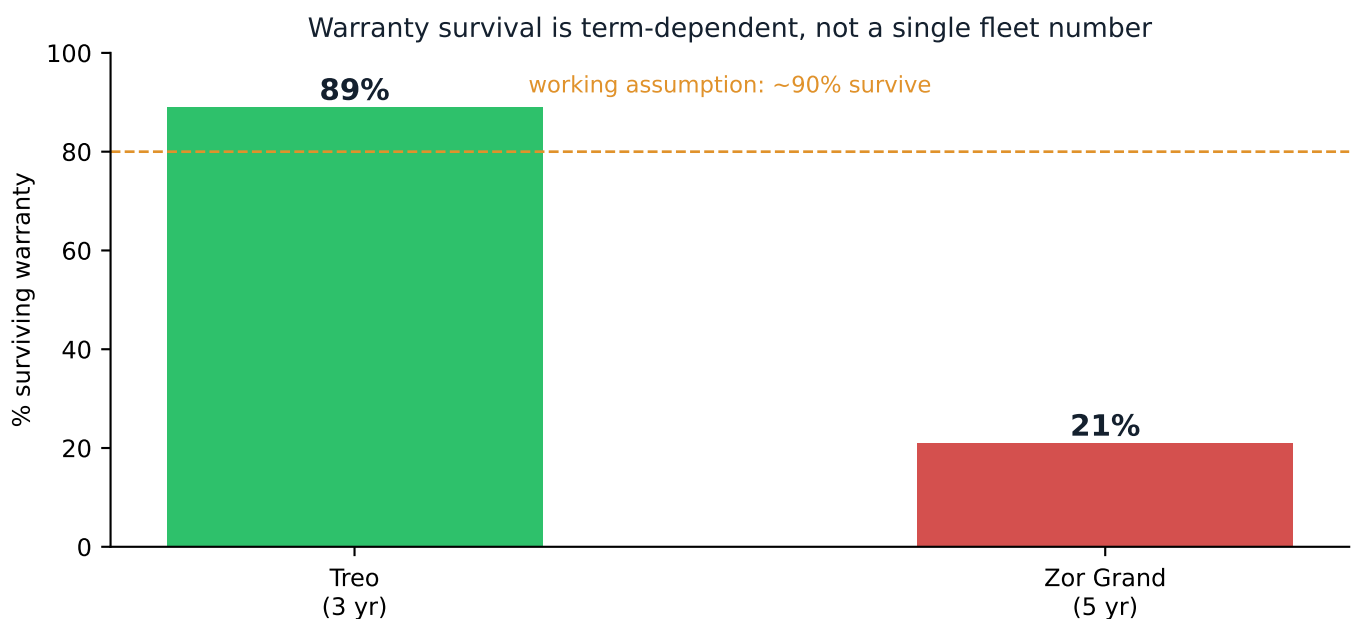
How to read it

Survival is reported per model/term. On short terms most vehicles pass; the long (5-yr) terms and the aged cohorts are where exposure concentrates.

Mahindra — warranty survival by model/term

Free-run SoH to each vehicle's warranty boundary; survive = projected SoH $\geq$ 80% at expiry.

Model	Vehicles	Warranty	Survive $\geq$ 80%	At-risk
Treo	79	3 yr	89%	9
Zor Grand	14	5 yr	21%	11

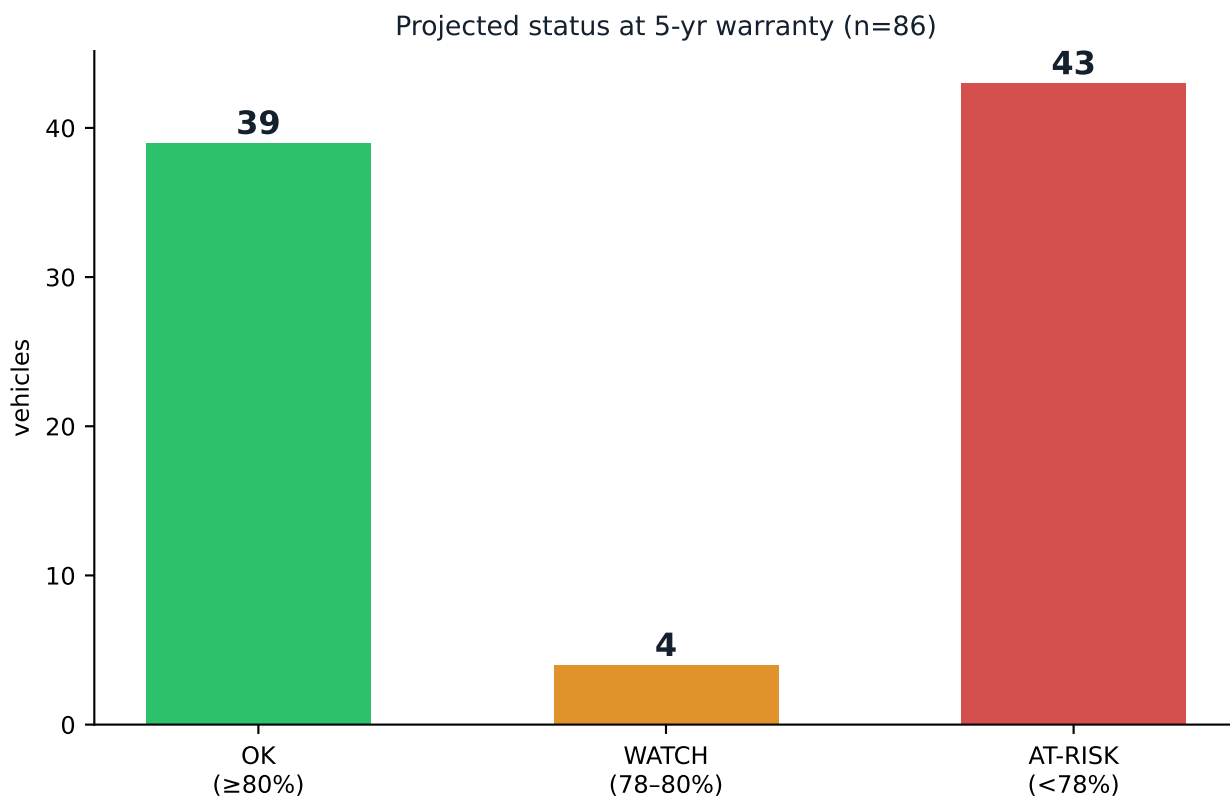


Key findings

- Treo (3-yr term, n=79): ~89% survive — the working ' ~90% survive ' assumption holds; short term is easy.
- Zor Grand (5-yr term, n=14): only ~21% survive — the 5-year term is where the real liability sits.
- Fleet-wide 78% survive (81% among the high-confidence $\geq$ 8-month vehicles).
- km is NEVER the binding limit (0/95) — low utilisation means calendar/condition aging drives risk.
- Action: split warranty exposure by model/term; the Zor Grand 5-yr cohort is the one to provision for.

Euler — 5-year warranty exposure

BMS-capacity SoH forecast to the 5-yr boundary; this is an aged resale cohort.

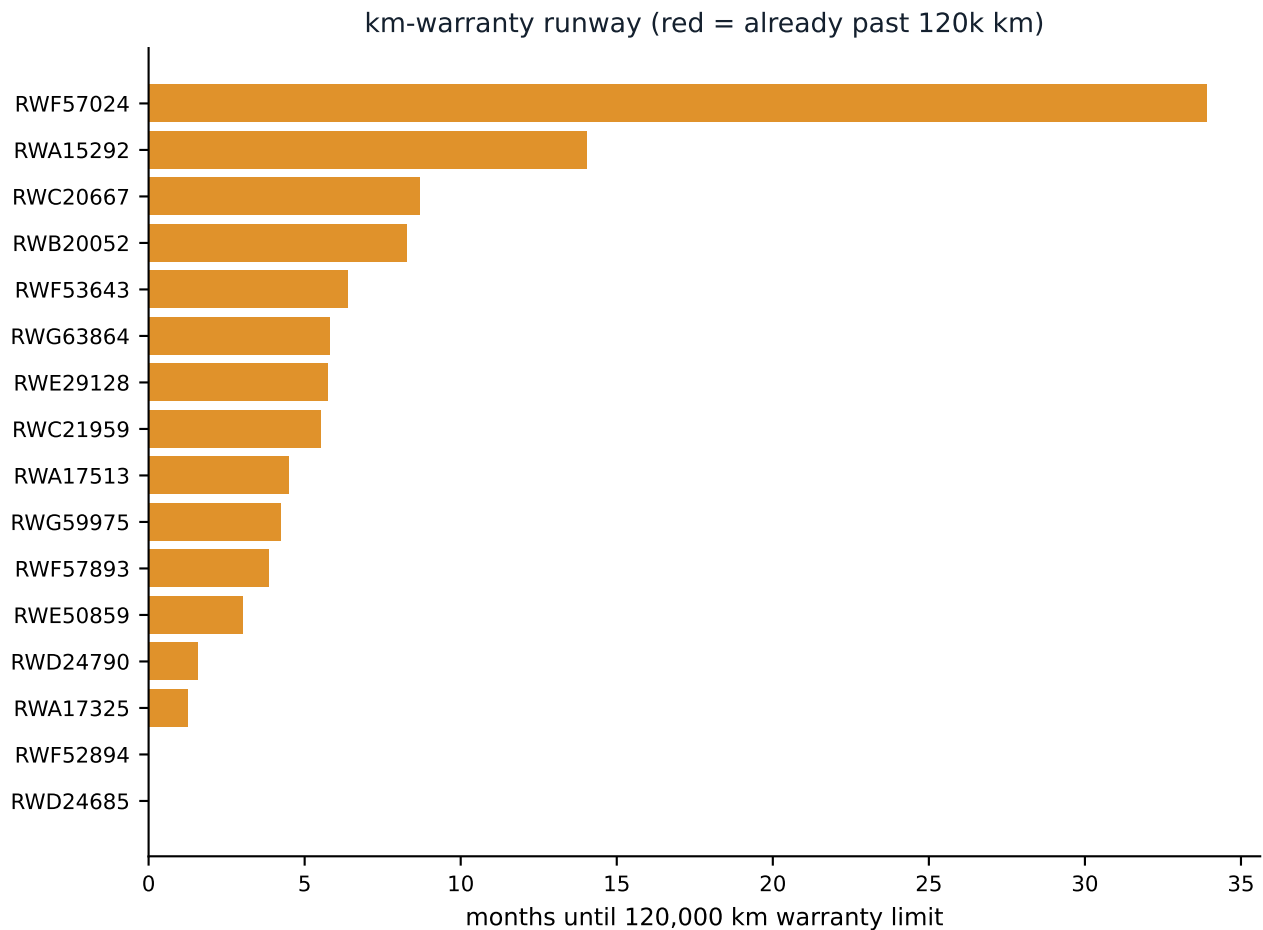


Key findings

- Only ~45% (39/86) are projected to survive the 5-yr warranty above 80% SoH.
- 43 vehicles AT-RISK, 4 WATCH — high exposure, but expected: this is an aged resale cohort.
- Among well-observed vehicles (≥8 months), ~48% survive.
- Caveat: 5-yr term + already-degraded fleet ⇒ the survival rate is far lower than a fresh fleet would show; read it as exposure on an old cohort, not a build-quality verdict.
- Action: prioritise these for resale/second-life routing before SoH erodes resale value further.

Bajaj — km-warranty runway & cycle RUL

High-mileage cargo fleet with a real BMS cycle counter. Here the km limit binds — the opposite of Mahindra.



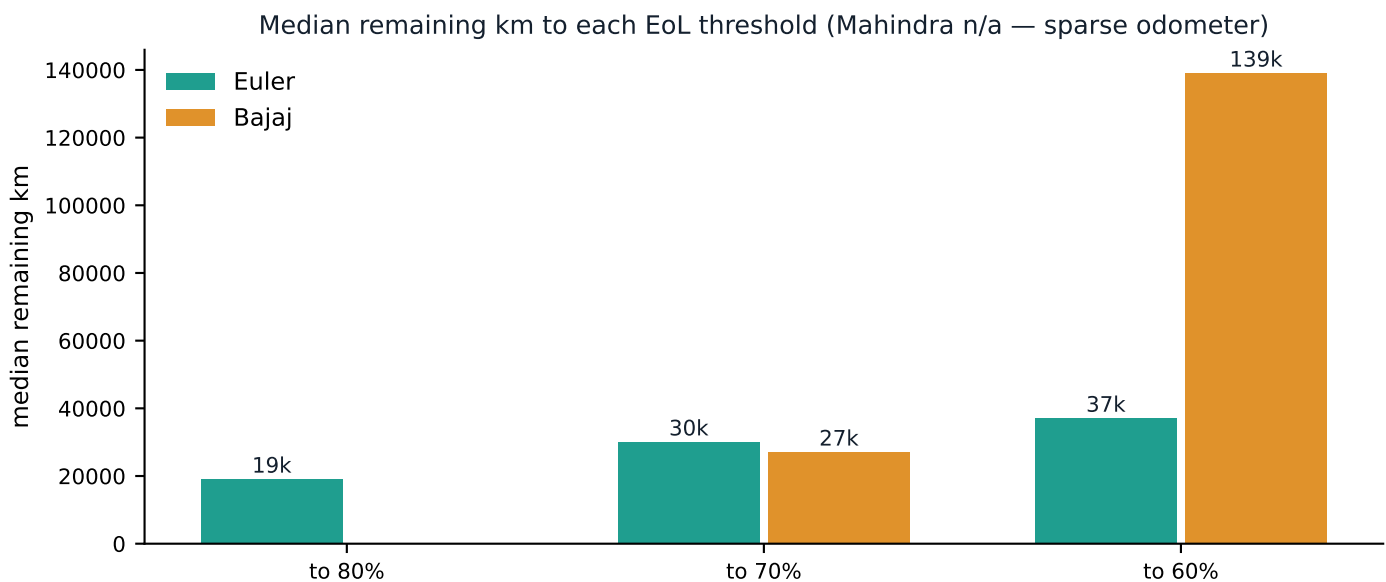
Key findings

- 2/16 vehicles have ALREADY exceeded the 120,000 km warranty limit.
- Usage is heavy: 1821–5438 km/month; the rest reach 120k in a median ~5.6 months.
- Cycle-based RUL: at current usage, ~3 months / ~138 more charge cycles to 70% SoH (median).
- For Bajaj the km limit is the binding constraint — opposite of the Mahindra fleet where time/SoH binds.
- Action: km-warranty provisioning matters here; flag the units already past 120k.

Remaining distance to End-of-Life (RUL in km)

remaining km = recent km/month × months until the SoH forecast reaches the EoL threshold.

OEM	km/month	to 80% (EoFL)	to 70%	to 60% (EoL)
Euler	1,500	19,000	30,000	37,000
Bajaj	4,160	n/a	27,000	139,000
Mahindra	n/a	n/a	n/a	n/a



Why km, and the caveats

- Degradation is calendar/condition-driven, not mileage-driven — so the battery ages out over TIME and km accrue at the usage rate. A high-utilisation vehicle therefore delivers MORE total km before the same SoH.
- EoL thresholds: 80% = end of first life (second-life trigger); 70%; 60% = true end of life.
- Mahindra: remaining-km is n/a for most vehicles — the odometer is logged too sparsely to derive km/month.
- Long extrapolations (to 60%) are low-confidence, especially Bajaj (√t fit far past its ~10-month window).
- 'Beyond horizon' (in the dashboard) means the vehicle does not reach the threshold within the modelled window.

Source: src/rul_km.py (km/month × months-to-EoL from the per-OEM SoH forecast). Surfaced live in the dashboard.

Summary

Where the warranty exposure sits

Headline

- Mahindra Treo (3 yr): ~89% survive — assumption holds. Zor Grand (5 yr): ~21% survive — the liability.
- Euler (5 yr, aged resale cohort): only ~45% projected to survive above 80%.
- Bajaj (high-mileage): 2/16 already past the 120k km limit; km binds, not time.

The cross-cutting lesson

- Warranty exposure is term- and usage-dependent — never a single fleet percentage.
- Low-mileage fleets (Mahindra) are bound by the time/SoH limit; km essentially never binds.
- High-mileage fleets (Bajaj) are bound by the km limit; SoH is secondary.

Actions

- Provision warranty reserves by model/term — concentrate on Mahindra Zor Grand (5 yr) and the Euler cohort.
- For Bajaj, track the km limit and flag units already past 120k.
- Exclude/flag low-confidence vehicles (<8 mo history) before making firm calls.